ENGINEERING DRAWING WITH AUTOCAD

UNIT 3: ORTHOGRAPHIC PROJECTIONS

Curriculum: Lovely Professional University | School of Computer Science & Engineering
Reference: Engineering Drawing with an Introduction to AutoCAD by Dhananjay Jolhe

>> 3.1 Principles and Geometry of Multiview Projection

Orthographic projection projects parallel perpendicular sight rays from an object to mutually perpendicular viewing planes:
* The Three Principal Views: Front View (Elevation), Top View (Plan), and Side View (Profile view / End view).
* Glass Box Concept: The object is imagined inside an imaginary glass box. Projections of features onto the 6 walls are folded flat into a 2D plane.

>> 3.2 First Angle vs Third Angle Projection Systems

* 1. First Angle Projection (ISO / Indian BIS Standard):
- Object is placed between Observer and Projection Plane (Quadrant 1).
- Front View is ABOVE the Top View.
- Left Side View is drawn on the RIGHT of Front View.
- Right Side View is drawn on the LEFT of Front View.
- Symbol: Truncated cone with small circle on right and front view trapezoid on left.
* 2. Third Angle Projection (American ANSI Standard):
- Projection Plane is placed between Observer and Object (Quadrant 3, plane assumed transparent).
- Top View is ABOVE the Front View.
- Left Side View is drawn on the LEFT of Front View.
- Right Side View is drawn on the RIGHT of Front View.
- Symbol: Truncated cone with small circle on left and front view trapezoid on right.
* Comparative Summary Table:
- Parameter | First Angle System | Third Angle System
- Plane Transparency | Opaque planes behind object | Transparent planes in front of object
- View Layout | Inverted (Left on Right, Top below) | Natural (Left on Left, Top above)
- Standard Code | BIS SP 46, ISO 128 | ASME Y14.3M

>> 3.3 Rules for Projecting Multiviews

* Alignment Principle: Front and Top views must be vertically aligned along common projector lines. Front and Side views must be horizontally aligned.
* Line Precedence: Solid visible outline > Dashed hidden line > Cutting plane line > Center line > Extension/dimension line.
* Miter Line Technique: A 45-degree line drawn from the intersection of projection axes allows seamless transfer of depth dimensions between Top and Side views without manual measurement.

>> 3.4 AutoCAD 2D Modification and Precision Transformation Commands

* LINETYPE (LT): Loads dashed, center, hidden linetypes. LTSCALE controls global scale factor of dash lengths.
* MOVE (M) & COPY (CO): Translates selected objects by base point and displacement vector.
* ROTATE (RO): Rotates entities about specified base point by given angle or reference angle.
* TRIM (TR) & EXTEND (EX): Trims objects at cutting edges or extends objects to boundary edges.
* ERASE (E): Removes selected entities from database.
* MIRROR (MI): Reflects objects across specified mirror line; option to delete source object or retain it.
* SCALE (SC): Enlarges or reduces objects proportionally by a numerical scale factor about base point.
* FILLET (F): Connects two non-parallel lines with a smooth tangent arc of specified radius (R=0 creates sharp corner).
* CHAMFER (CHA): Creates beveled edge between two lines defined by distance1 and distance2 or distance and angle.
* ARRAY (AR): Generates associative repeating patterns: Rectangular Array (rows, columns, spacing) or Polar Array (items, angle to fill, center point).

[PROBLEM 1: 3-View First Angle Projection of Stepped Block]

Problem Statement: An isometric block has base 80 mm x 50 mm, total height 45 mm. It has a step of height 20 mm extending 35 mm from the left edge. A central through rectangular slot of 20 mm width x 15 mm depth runs across the entire 50 mm depth on the upper surface. Draw Front View, Top View, and Left Side View in First Angle Projection.

Step-by-Step Multiview Construction:
* 1. Front View (Looking from Front, showing Length 80 mm and Height 45 mm):
- Base line = 80 mm. Left vertical rise = 20 mm. Horizontal step = 35 mm. Vertical rise to top = 25 mm (reaching total 45 mm).
- Right edge = 45 mm vertical. Top horizontal width = 80 - 35 = 45 mm.
- Central slot: Width 20 mm, depth 15 mm. Drawn as visible solid cutout on the top surface between x = 47.5 mm and 67.5 mm.
* 2. Top View (Looking from Above, aligned vertically below Front View, showing Length 80 mm and Width 50 mm):
- Overall boundary rectangle = 80 mm x 50 mm.
- Step line: Visible solid line at x = 35 mm from left edge running full 50 mm width.
- Slot: Two parallel visible lines spaced 20 mm apart running horizontally across the right block portion.
* 3. Left Side View (Drawn on the RIGHT side of Front View as per First Angle rules):
- Shows Width 50 mm and Height 45 mm.
- Base width = 50 mm, left and right heights = 45 mm. Visible line at height 20 mm representing top edge of lower step.
- Slot shown as central cutout of width 20 mm and height 15 mm down from top edge.

[PROBLEM 2: Bracket with Circular Bore and Hidden Feature Handling]

Problem Statement: A cast iron bracket consists of an L-flange with vertical plate (thickness 15 mm, height 60 mm, width 50 mm) and horizontal base (thickness 15 mm, length 70 mm, width 50 mm). The vertical plate has a central through circular hole of 20 mm diameter located 35 mm above the base. Draw Front View and Top View.
Solution & Drafting Conventions:
* Front View: Shows L-profile. Base 70 mm x 15 mm, vertical arm 15 mm x 45 mm. The circular hole appears on the vertical plate: center mark at height 15 + 35 = 50 mm from bottom, diameter 20 mm drawn as visible circle with horizontal and vertical centerlines (chain thin lines).
* Top View: Placed directly below Front View. Overall rectangle 70 mm x 50 mm. The vertical plate appears as a 15 mm x 50 mm rectangle on the left. The circular hole cannot be seen from above: it passes through the 15 mm thickness. It is represented by two dashed hidden lines spaced 20 mm apart, with a chain-thin centerline running midway.

[PROBLEM 3: Conversion of 3D Bearing Bracket: 1st vs 3rd Angle Layout]

Problem Statement: Compare the placement and projection orientation of a bearing bracket in First Angle versus Third Angle projection systems.
Detailed Comparison Matrix & View Layouts:
* 1. First Angle System Layout:
- Top-Left: Right Side View (looking from right of bracket).
- Center-Top: FRONT VIEW (Elevation).
- Top-Right: Left Side View (looking from left of bracket).
- Center-Bottom: TOP VIEW (Plan view, looking from above).
* 2. Third Angle System Layout:
- Center-Top: TOP VIEW (Plan view, looking from above).
- Top-Left: Left Side View (looking from left of bracket).
- Center-Bottom: FRONT VIEW (Elevation).
- Bottom-Right: Right Side View (looking from right of bracket).
* Engineering Conclusion: First angle projects views onto the far plane; third angle projects views onto the near transparent plane. Dimensions across views remain identical.

[PROBLEM 4: AutoCAD Multiview Drafting Protocol for Machine Support]

Problem Context: Draft the 2D orthographic views of a shaft support bracket using AutoCAD modification tools.
AutoCAD Command Step Log:
* 1. Layer Setup: Activate Layer "0_Visible".
* 2. RECTANG -> 0,0 to 80,45 (Front view perimeter).
* 3. FILLET -> Radius = 10 -> Select top-right corner edges to create smooth rounded shoulder.
* 4. CIRCLE -> Center at 40,25, Radius 12 mm for shaft bore.
* 5. MIRROR -> Select left reinforcement web -> Mirror line at X=40 to duplicate perfectly to the right side.
* 6. Project Top View: Draw construction lines downward using XLINE -> Vertical -> Snap to front view vertices.
* 7. OFFSET -> 25 mm below front view for top view base -> RECTANG 80 mm x 50 mm.
* 8. LTSCALE -> Set to 0.5 to ensure dashed hidden lines of the 24 mm hole display clearly in top view.